

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

Pilani Campus

ECON F311

International Economics

Does Trade Openness Encourage Clean Energy Investment?

Evidence from Emerging Markets

Group Number: 17

Student Name	BITS ID
Ritik Yadav	2024B3A30940
Sriansh Rajan	2024B3A80901P
Nidhish Kumar Tyagi	2024B3A80952P
Heet Jain	2024B3A80929P
Jaivardhan Singh	2024B3A40859P

Submitted to: Prof. Geetilaxmi Mohapatra

October 2026

Abstract

This study asks whether trade openness is associated with higher clean-energy investment in emerging markets. Because comparable observations of total realised investment are unavailable, the empirical outcome is a narrower measure: public renewable-energy finance commitments recorded by the International Renewable Energy Agency (IRENA). The sample is selected from the International Monetary Fund emerging and developing economy classification, with a pre-specified 2023 nominal GDP threshold of US$100 billion. Annual IRENA data are merged with World Bank indicators for 2014–2023. The corrected complete-case panel comprises 305 country-year observations in 35 economies. This panel data analysis estimates pooled ordinary least squares (OLS), country fixed effects, and country-and-year fixed effects with country-clustered standard errors. Trade openness is measured as exports plus imports relative to GDP; income and inflation are controls, while foreign direct investment is examined separately. Diagnostic testing and a transparent missing-data audit guide specification choices. In the two-way fixed-effects model, the trade coefficient is positive but small and statistically indistinguishable from zero (coefficient 0.0027; $p = 0.919$). Inflation is also statistically insignificant after correcting the treatment of missing finance values. The results therefore do not establish that more trade-intensive economies attract larger public renewable-finance commitments. They do not rule out meaningful effects on private finance, disbursements, installed capacity, or realised clean-energy investment. Policy implications emphasise combining technology access with credible renewable-energy project frameworks, investment conditions, and improved finance statistics rather than treating trade openness as an independently proven driver of clean-energy investment.

1. Introduction

Can international trade help emerging economies move towards cleaner energy? Access to imported equipment, foreign suppliers and new technology may lower the cost of renewable projects. Trade can also increase electricity demand, support fossil-intensive production and expose investment to external shocks. Both directions are plausible, so the answer depends on the evidence.

Trade can affect both project costs and energy demand. Imported equipment and specialist services may make solar, wind and grid projects cheaper to build. Export growth may raise electricity use, but the additional demand could be met by renewables or by existing fossil technologies. A higher trade-to-GDP ratio therefore does not necessarily mean lower tariffs or more renewable finance. The ratio reflects economic activity as well as policy.

This paper examines that relationship in emerging markets. A practical difficulty is that country-year data often measure renewable capacity, electricity generation or consumption rather than annual investment. These outcomes are not interchangeable. The analysis therefore uses IRENASTAT public renewable-energy finance commitments to capture the public-financing component of clean-energy investment, and sets out what this measure can and cannot tell us.

A public financing commitment is not the same as money spent on a completed project. Development banks, bilateral agencies and other public lenders may approve a large amount before construction or disbursement begins. Project readiness, recipient institutions and donor priorities can influence those decisions. Commitments may therefore follow a different pattern from private spending or installed capacity. The model examines this public-financing relationship, rather than the effect of openness on total realised investment.

The analysis brings together the literature on trade, technology and renewable energy, then combines IRENA finance data with World Bank indicators. Pooled OLS, country fixed-effects and two-way fixed-effects models are compared using clustered inference. The results are interpreted as conditional associations. In the corrected two-way model, the trade coefficient is positive but close to zero and estimated with considerable uncertainty. Neither trade nor inflation is statistically significant, so the results do not establish that greater openness independently raises public renewable-energy finance commitments.

1.1. Research gap, objectives and hypotheses

The literature has examined trade and renewable-energy consumption extensively, but a smaller literature directly evaluates investment or finance. The present study contributes an explicitly delimited outcome (public finance commitments), an IMF-based emerging-market country rule and reproducible missing-data controls. These choices enable a transparent associational test without claiming causal identification or novel access to total investment records.

The objectives are (i) to measure the relationship between trade openness and public renewable-finance commitments after accounting for income and inflation; (ii) to compare pooled and fixed-effects estimates and assess model diagnostics; and (iii) to evaluate sensitivity to potential financing channels such as foreign direct investment. The hypothesis is H_0: $\beta_1 = 0$ against the theory-motivated one-sided H_1: $\beta_1 > 0$, while two-sided intervals and p-values are reported for caution. Positive technology-diffusion effects compete with energy-demand and fossil-production channels, so the sign is not assumed.

The null and alternative are stated for the trade coefficient in the country-and-year effects specification. Reporting two-sided confidence intervals alongside a theory-motivated directional hypothesis prevents a positive point estimate from being mistaken for decisive support when its uncertainty is substantial. The paper also asks whether a result is economically informative: a null-hypothesis test is only one component of the assessment.

2. Literature Review

2.1. Trade, technology and competing effects

Theoretical work emphasises several channels through which trade can affect environmental and energy outcomes. Trade may improve allocative efficiency and facilitate technology diffusion, but it may also expand pollution-intensive activity depending on comparative advantage and regulation (Antweiler et al., 2001; Keller, 2004). In the renewable-energy context, imports of turbines, photovoltaic modules and related balance-of-system components can reduce project costs and broaden access to technology, especially in economies with limited domestic manufacturing capability.

These mechanisms can be understood as scale, composition and technique effects. A scale effect arises when expanded trade increases production and energy use. A composition effect appears when trade shifts activity towards manufacturing, extraction or services with different energy requirements. A technique effect arises through improved equipment, learning and regulation. The same increase in measured openness can produce competing effects, especially when the renewable project pipeline reacts slowly. Antweiler et al. (2001) develop this broader trade-and-environment framework; here it motivates competing predictions rather than a causal decomposition estimated from finance data.

2.2. Evidence on renewable-energy outcomes

Empirical evidence for renewable-energy outcomes is mixed. Murshed (2020) examines low- and middle-income countries and finds that trade liberalisation can support renewable-energy transition, but the effects vary by income group and the dependent variable is renewable-energy use rather than investment. Zeren and Akkuş (2020) find long-run and directional linkages between renewable and non-renewable energy consumption and trade openness, including feedback in which renewable use may reduce trade intensity. Zhang et al. (2021) identify nonlinear effects of trade measures on

renewable consumption in OECD economies. These studies are useful but do not directly answer an investment question.

The distinction among renewable-energy consumption, installed generating capacity and finance is central. An imported turbine can expand installed capacity only after planning and construction; a commitment may be recorded before any import occurs. Conversely, rising electricity demand can reduce the renewable share even while financing or renewable generation increases in absolute terms. Consequently, a negative coefficient in a consumption-share study does not automatically contradict a positive coefficient in a financing study, and the studies cannot be pooled as though they measured the same phenomenon.

2.3. Investment, governance and foreign capital

A closer precedent is Hussain et al. (2021), who study renewable-energy investment in Belt and Road Initiative countries and find that governance and trade openness matter for renewable investment outcomes. Yet their sample, institutional context and design differ from a broader emerging-market focus. Kim and Kim (2015) highlight the interaction between policy, innovation and international trade in renewable-energy technologies, reinforcing the argument that trade can support clean-energy deployment through knowledge and equipment channels. The literature therefore suggests plausible positive mechanisms, but it does not imply a universal or necessarily causal positive effect in all emerging markets.

Foreign direct investment has an ambiguous empirical role. Trade integration may encourage multinational activity and project financing, making FDI a possible mediator from openness to renewable investment. Yet general FDI statistics include fossil extraction, manufacturing and unrelated services. Including FDI as a baseline control could therefore mask part of the pathway of interest, while excluding it may leave residual confounding from broader financial conditions. This paper uses FDI as a disclosed sensitivity specification rather than presenting its coefficient as a structural clean-energy-finance effect.

2.4. Synthesis and research contribution

The study contributes in two ways. It uses a stated IMF classification and economic-size rule to select emerging markets, rather than choosing countries without a clear rule. It also distinguishes public renewable-finance commitments from wider measures of clean-energy investment. This makes the outcome easier to interpret, although it narrows the question the data can answer.

A statistical association is not automatically a policy effect. Country characteristics, global shocks, financing cycles and selective data coverage can affect both trade and recorded commitments. Country and year effects address some of these differences; confidence intervals and sensitivity checks show how much uncertainty remains. They do not replace an external instrument or a credible

trade-policy experiment. The contribution is a transparent analysis of an investment-related measure, rather than a claim to identify causation.

Table 1. Selected empirical literature relevant to the study

Study	Sample and period	Outcome and method	Main relevance
Murshed (2020)	71 low- and middle-income economies, 2000–2017	Renewable-energy transition; IV panel	Shows that trade-related policies can matter, but outcome is not investment.
Zeren & Akkuş (2020)	Bloomberg top emerging economies; 1980–2015	Renewable-energy consumption; cointegration and causality	Reports long-run/directional linkages, not a simple positive investment effect.
Zhang et al. (2021)	35 OECD countries, 1999–2018	Renewable-energy consumption; panel smooth transition	Highlights possible nonlinearities and limited external validity for EMDEs.
Hussain et al. (2021)	51 BRI countries, 1996–2017	Renewable-energy investment; panel methods	Closest investment-oriented precedent; shows relevance of trade openness and governance.
Kim & Kim (2015)	International renewable technology panels	Technology trade and innovation	Supports the technology-diffusion mechanism behind the hypothesis.

Note. This table synthesises the role of prior work in motivating the present design rather than reproducing each study in full detail.

3. Data and Methodology

3.1. Country sample, period and data sources

The starting country universe is the April 2025 IMF World Economic Outlook list of emerging and developing economies. A transparent economic-size rule then retains economies with nominal GDP of at least US$100 billion in 2023 using WDI indicator NY.GDP.MKTP.CD. This yields 43 eligible economies. The study period is 2014–2023, chosen to balance recent relevance and data availability in both IRENA and WDI. After merging the datasets and dropping observations with missing baseline regressors, the final complete-case sample contains 305 country-year observations across 35 countries.

Applying the GDP threshold to the IMF list produces 43 eligible countries and 430 possible country-years. IRENA contains non-missing constructed renewable-finance entries for 321 of those country-years (36 countries); a further restriction to observed trade openness, income and inflation produces 305 complete country-years in 35 countries. The panel is therefore unbalanced. Country-year rows with no recorded finance amount are not treated as zero, and estimates apply only to this observed complete-case subset. The record count is not evidence that all eligible countries had projects, nor is a missing cell evidence of no investment.

The dependent variable is constructed from IRENASTAT “Public Investments (2023 Million USD) by Country/area, Technology and Year.” This is best interpreted as public renewable-energy finance commitments in constant 2023 million USD, not as total realised clean-energy investment. Trade openness is measured as total trade in goods and services as a share of GDP. Real GDP per capita captures income and absorptive capacity, while CPI inflation proxies macroeconomic instability. FDI inflows are used only in a robustness check because they may partly mediate the effect of trade openness.

Table 2. Variables, definitions and sources

Variable	Definition and transformation	Source / code
Public renewable-finance commitments	IRENA public renewable finance commitments, 2023 USD million; asinh transformation.	IRENASTAT PUBFIN table
Trade openness	(Goods/service exports + imports) ÷ GDP × 100; trade intensity.	World Bank WDI, NE.TRD.GNFS.ZS
GDP per capita	ln(real GDP per capita), constant 2015 US dollars.	World Bank WDI, NY.GDP.PCAP.KD
Inflation	Consumer price inflation, annual percent.	World Bank WDI, FP.CPI.TOTL.ZG
FDI inflows (robustness)	Net FDI inflows, % of GDP; sensitivity control.	World Bank WDI, BX.KLT.DINV. WD.GD.ZS

Note. The outcome is a proxy for the public-financing margin of renewable investment, not a measure of total clean-energy capital expenditure.

Trade openness is WDI indicator NE.TRD.GNFS.ZS, measured in percentage points of GDP, so β_1 is per additional percentage point rather than per unit interval. Income is the natural logarithm of real GDP per capita; its coefficient captures a change associated with proportional income differences. Inflation is the annual CPI percentage-change indicator; its inclusion reflects financing uncertainty, although inflation may also respond to policy or commodity shocks. Using a compact control set limits degrees-of-freedom pressure in a ten-year panel and makes the mediator-versus-confounder distinction for FDI explicit.

3.2. Construction of the dependent variable

The IRENA export contains twenty technology labels but no verified all-renewables total row. The study excludes coal, oil, gas, other fossil fuels, nuclear energy, non-renewable waste and mechanical storage from its renewable proxy. It sums ten renewable labels (hydropower, photovoltaic and solar thermal, geothermal, marine, renewable waste, multiple renewables and three biofuel classes). Because the taxonomy distinguishes generic “Wind energy” from onshore and offshore wind, the

conservative wind contribution is the maximum of the generic wind entry and the onshore-plus-offshore sum, using only genuinely observed entries. This avoids automatically adding potential parent/subcategory values, but may undercount finance if the wind labels refer to different projects. IRENA does not provide project-level records in this export with which to verify that distinction. The code and limitation note disclose this construction explicitly.

Finance totals are calculated in 2023-price millions of US dollars, using IRENA's published unit convention. The technology categories are kept at their source definitions. In particular, "Multiple renewables" is retained as a distinct category, but the source lacks project identifiers to verify whether every technology label is disjoint. The main aggregate thus requires a documented classification judgement. A separate additive-wind specification in the results section shows whether that judgement materially changes the trade estimate.

Missing values (the IRENA symbol "..") are kept missing, including during wind aggregation. An outcome is assigned zero only when at least one included component is an explicitly observed zero and no included component is positive. All-missing country-years remain missing. This treatment is essential because indiscriminate zero filling can create artificial country-year observations and change the estimated regression sample.

3.3. Econometric specification

The baseline estimating equation is:

$$\operatorname{asinh}(C_{it}) = \beta_0 + \beta_1\, T_{it} + \beta_2 \ln(GDPpc_{it}) + \beta_3\, \pi_{it} + \alpha_i + \lambda_t + \varepsilon_{it} \quad (1)$$

In Equation (1), C_{it} denotes public renewable-finance commitments in constant 2023 USD millions for country i in year t; T_{it} is trade openness in percentage points of GDP; $GDPpc_{it}$ is real GDP per capita; and π_{it} is annual CPI inflation in percent. The terms α_i and λ_t are country and year intercept effects, while ε_{it} captures disturbances. All β parameters are conditional associations; β_1 is the coefficient of direct interest. Pooled OLS omits both effect sets, the country-fixed-effects model retains α_i, and the two-way model includes both.

$$T_{it} = 100 \times (X_{it} + M_{it}) \,/\, GDP_{it} \quad (2)$$

Equation (2) defines the World Bank trade ratio using exports X and imports M of goods and services. This ratio can exceed 100% for economies with large trade flows relative to measured output; it does not measure tariff rates or non-tariff barriers directly.

$$\operatorname{asinh}(C_{it}) = \ln\left(C_{it} + \sqrt{(C_{it}^2 + 1)}\right) \quad (3)$$

Equation (3) specifies the transformation in the numeric unit of millions of 2023 USD. It is defined for a genuinely observed zero and reduces skewness, but its regression coefficient is not a constant percentage effect of finance at all magnitudes. A change in the underlying monetary unit would also

change the transformed scale; the unit is held fixed across observations. Missing finance values remain missing throughout the transformation.

Pooled OLS is reported as a benchmark. The main models add country fixed effects and then both country and year fixed effects. Country fixed effects absorb time-invariant heterogeneity such as geography, long-run institutions and resource endowments, while year fixed effects absorb common shocks. Because the outcome includes zeros and is strongly right-skewed, the inverse hyperbolic sine transformation is preferred to a log transformation. Standard errors are clustered at the country level.

The country fixed-effects specification identifies β_1 using within-country variation, rather than permanent differences in geography, donor access or industrial structure. Year effects allow for global renewable technology price trends, commodity-market fluctuations and changes in international public financing. These terms are retained for economic reasons even if a particular significance test is borderline. Random effects would require the country-specific component to be uncorrelated with trade and the controls, an implausibly strong requirement in this setting; it is therefore not selected merely for computational convenience.

The primary hypothesis is H_0: $\beta_1 = 0$ against H_1: $\beta_1 > 0$. Nevertheless, because endogeneity and omitted variables remain possible, the empirical results are interpreted as associations rather than causal effects.

3.4. Diagnostic strategy and model selection

The pooled Breusch–Pagan test examines homoskedasticity; the joint country and year effect tests assess whether the corresponding intercept sets improve the nested least-squares fit (these classical F tests are not cluster-robust). Residual cross-sectional correlation is screened with an approximate Pesaran CD calculation, while within-country variance inflation factors screen multicollinearity. A difference-residual lag test is reported as exploratory only and must not be confused with a standard Wooldridge panel test. Regardless of these checks, country-clustered standard errors are used because commitment shocks can persist within economies. With 35 country clusters, asymptotic inference is informative but not exact. Year effects are retained to address common global shocks, and the final two-way model is justified by the panel structure rather than a preferred coefficient sign or fit statistic.

Tests for panel unit roots and cointegration are not automatically transferable from a long, relatively complete consumption panel to this shorter, irregularly observed commitments panel. The ten-year window, numerous missing outcomes and lumpy public project approvals reduce the reliability of conventional panel integration tests. We therefore avoid claiming stationarity or cointegration results that were not established, and do not impose an ARDL or error-correction interpretation. Equation (1) estimates a conditional levels association, with this time-series limitation disclosed.

Model selection is based on research design and residual behaviour, not on whichever specification gives the largest adjusted R^2. The pooled model provides an unconditional benchmark, while joint

effect tests assess whether the common-intercept restrictions are supported. The Breusch–Pagan test concerns pooled-model heteroskedasticity; a non-significant result would not prove that panel errors are independent. Clustering by recipient economy allows arbitrary within-country heteroskedasticity and serial dependence, but may still be imprecise with 35 country clusters. The serial-correlation and cross-sectional-dependence calculations in this project are exploratory approximations and are labelled accordingly in Table 7.

Table 3. Sample construction summary

Step	Value
Eligible EMDEs (GDP>=US$100bn, 2023)	43
Eligible EMDEs with nonmissing outcome in panel window	36
Countries in complete-case baseline regressions	35
Country-year observations in full eligible grid	430
Country-year observations with nonmissing outcome	321
Country-year observations in baseline regressions	305

Note. The final baseline sample is a complete-case panel for the baseline specification only.

4. Results and Discussion

4.1. Descriptive evidence

The complete-case baseline panel covers 305 observations from 35 countries: AGO, ARE, ARG, BGD, BRA, CHL, CHN, COL, DOM, DZA, ECU, EGY, ETH, GTM, IDN, IND, IRN, IRQ, KAZ, KEN, MAR, MEX, MYS, PAK, PER, PHL, QAT, SAU, THA, TUR, UKR, UZB, VEN, VNM, ZAF. Within this complete-case panel, mean reported public renewable-finance commitments are about 422.8 million 2023 USD, but the large standard deviation and maximum value indicate substantial skewness. Trade openness averages 60.0 percent of GDP, while average inflation is 9.1 percent.

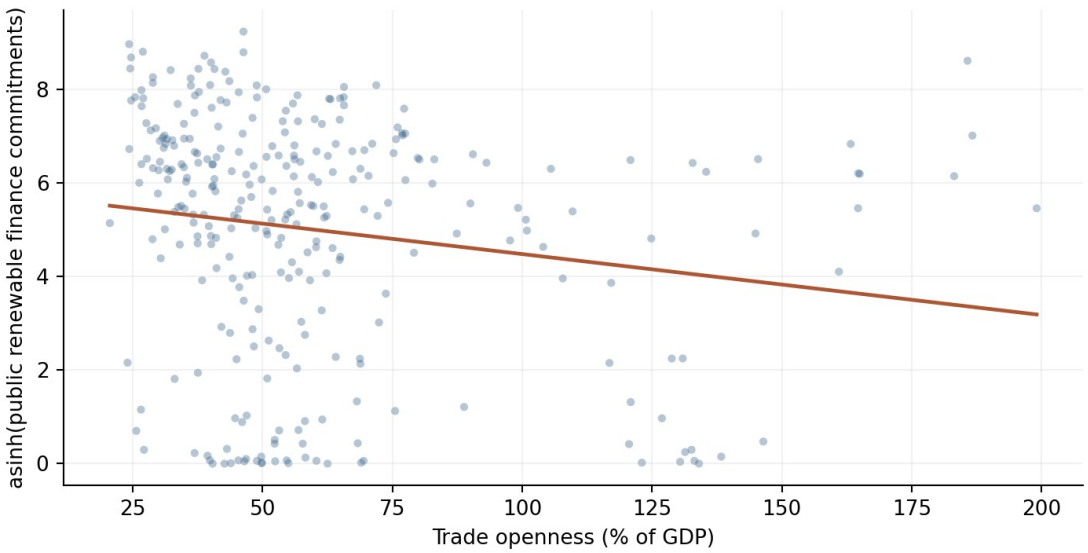
Recorded commitments are highly right-skewed: the observed median is US$132.28 million and the maximum is US$5,145.91 million, compared with a mean of US$422.77 million. This is one reason for using asinh rather than untransformed monetary levels in the main model. Trade openness averages 59.96% of GDP in the same complete-case sample. These descriptive quantities use precisely the country-years that enter the baseline regressions; they should not be generalised to all 43 eligible economies.

Table 4. Descriptive statistics (baseline complete-case sample)

Variable	Obs.	Mean	Std. dev.	Min	Median	Max
public_commitments_musd2023	305	422.77	700.38	0.00	132.28	5145.91
asinh_commitment	305	5.00	2.55	0.00	5.58	9.24
trade_pct_gdp	305	59.96	33.13	20.60	50.91	199.04
ln_gdp_pc	305	8.44	0.77	6.32	8.45	10.96
inflation_cpi_pct	305	9.08	14.46	-2.54	5.13	133.49

Note. Public commitments are measured in constant 2023 million USD before the asinh transformation.

Figure 1. Trade openness and public renewable finance: pooled observations



Note. 305 observed country-years from the baseline sample. The fitted line is an uncontrolled pooled linear association; it is not the fixed-effects estimate or a causal relationship. Source: authors' calculations from IRENASTAT and World Bank WDI.

Figure 1 illustrates that country-years with similar trade-to-GDP ratios can have very different public commitment outcomes. The pooled slope is descriptive and can differ from the within-country coefficient once persistent national characteristics and common year shocks are absorbed. This distinction explains why a simple scatterplot alone cannot settle the research hypothesis.

Table 5. Correlation matrix

Variable	asinh(C_{it})	T_{it}	ln($GDPpc_{it}$)	π_{it}
asinh(C_{it})	1.000	-0.169	-0.064	-0.008
T_{it}	-0.169	1.000	0.242	-0.150
ln($GDPpc_{it}$)	-0.064	0.242	1.000	0.008
π_{it}	-0.008	-0.150	0.008	1.000

Note. Correlations are simple pairwise correlations within the baseline complete-case sample.

The correlation matrix similarly describes unconditional levels rather than controlled within-country movements. Cross-country income and industrial structure can generate correlations that change after fixed effects are introduced. Correlation coefficients are therefore useful for screening variable relationships and extreme collinearity, not for estimating a causal trade response.

4.2. Baseline regressions

Table 6 reports the corrected regression results. Trade openness is negative and insignificant in the pooled benchmark; after country fixed effects it remains very close to zero and is also insignificant. In the preferred two-way fixed-effects model its coefficient is 0.0027 (country-clustered SE 0.0268; p = 0.919; 95% CI −0.0517 to 0.0571). This interval includes meaningful positive and negative

associations, so the study cannot infer a robust relationship between trade intensity and public renewable-finance commitments. A one-percentage-point trade increase corresponds mechanically to 0.0027 asinh units, not a reliably estimated percentage change in total investment.

The substantive uncertainty is substantial. For illustration, a ten-percentage-point rise in trade openness mechanically maps into about 0.027 asinh units under the point estimate, while the corresponding 95% interval is approximately −0.517 to 0.571 asinh units. These are transformed-outcome associations, not percentage-point changes in renewable generation or percentage increases in total clean-energy capital investment. A statistically non-significant coefficient should not be interpreted as evidence that the true association is identically zero.

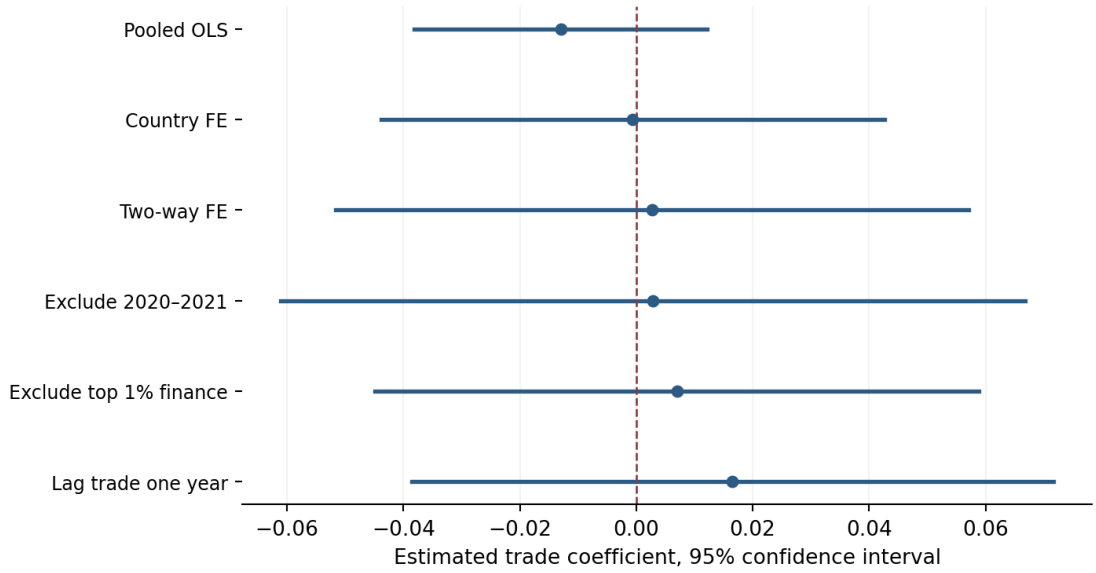
Unlike the earlier, incorrectly constructed panel, inflation is not statistically significant in the corrected two-way model (coefficient 0.0012; p = 0.913). Income has a positive coefficient but its two-sided p-value is 0.066, above the conventional 5% threshold. The estimated conditional patterns are therefore inconclusive. The country and year terms account for important heterogeneity but cannot eliminate endogeneity, donor-allocation confounding or technology-specific selection.

Table 6. Baseline regression results

Variable	(1) Pooled OLS	(2) Country FE	(3) Two-way FE
Trade openness	-0.013 (0.012)	-0.001 (0.021)	0.003 (0.027)
ln(GDP per capita)	-0.077 (0.354)	2.372 (1.641)	3.591* (1.892)
Inflation	-0.006 (0.014)	-0.001 (0.011)	0.001 (0.011)
Country FE	No	Yes	Yes
Year FE	No	No	Yes
Observations	305	305	305
Country clusters	35	35	35
Adjusted R-squared	0.021	0.589	0.602

Note. Entries are coefficients with country-clustered standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Figure 2. Trade-openness estimates across baseline and sensitivity models



Note. Dots are estimated coefficients and horizontal segments are 95% country-clustered confidence intervals. The vertical dashed line denotes zero. Lagged-trade, COVID-exclusion and top-finance-exclusion specifications have different samples; comparability is therefore conditional.

The estimates in Figure 2 remain centred near zero across the country fixed-effects and two-way models and the reported alternative samples. The pooled coefficient is somewhat more negative, again indicating that a between-country correlation need not describe the conditional within-country association. All displayed confidence intervals cross zero.

4.3. Diagnostic tests and robustness

The diagnostic table reports computed statistics rather than proposed tests. The pooled Breusch–Pagan result ($p = 0.244$) does not reject homoskedasticity in that benchmark; clustering is retained as a panel-design precaution, not claimed to be required by this test. Classical nested-model F tests indicate substantial country heterogeneity ($p < 0.001$) and year heterogeneity conditional on country effects ($p = 0.043$), although their non-robust p-values require caution. The approximate Pesaran CD statistic does not reject cross-sectional independence ($p = 0.143$). The exploratory residual-difference check ($p = 0.956$) cannot substitute for a fully implemented Wooldridge test. The maximum within-country VIF is 1.06, suggesting limited regressor collinearity.

The tests address different restrictions. The country-effect comparison rejects the restriction that all country intercepts are identical, reinforcing the concern that pooled regression mixes incomparable economies. The year-effect comparison also favours allowing common shocks, though its reported F test is conventional rather than cluster-robust. The limited VIF measures indicate that the baseline regressors do not have strong linear collinearity after within-country demeaning. None of these tests validates exogeneity of the openness regressor.

Adding FDI to the two-way model leaves the trade coefficient statistically indistinguishable from zero (−0.0013; p = 0.959). Inflation also remains insignificant (p = 0.868). The FDI coefficient is positive and marginal at the 10% threshold (p = 0.085), but it does not establish a direct financing mechanism: FDI may itself be jointly determined with investment commitments. The robustness check therefore reinforces the inconclusive trade result rather than a positive claim.

Table 7. Diagnostic-test results

Test	Null hypothesis	Stat.	p-value	Interpretation
Breusch-Pagan heteroskedasticity (pooled benchmark)	Homoskedastic pooled OLS residuals	4.168	0.244	Do not reject homoskedasticity in pooled benchmark; clustered inference retained for panel structure.
Exploratory residual-difference serial-correlation check	Differenced residual lag coefficient = -0.5	0.003	0.956	Exploratory, not a formal Wooldridge panel test; retain country-clustered SE.
Pesaran CD (two-way FE residuals)	Cross-sectional independence	-1.465	0.143	No rejection; the unbalanced-panel CD approximation is exploratory.
Joint country fixed effects F-test	All country effects equal zero	13.266	<.001	Classical F test supports country FE relative to pooled OLS (not cluster-robust).
Joint year fixed effects F-test	All year effects equal zero (conditional on country FE)	1.971	0.043	Classical F test rejects equal year effects at 5%; year FE also motivated by common shocks.
Max within-transformed VIF	No formal null (diagnostic threshold)	1.060		No evidence of problematic baseline regressor collinearity.

Note. Exploratory residual-difference statistic is a squared t statistic testing lag coefficient = −0.5; it is not a formal Wooldridge test. The fixed-effects nested F tests use conventional OLS covariance.

Table 8. Robustness check: two-way FE with FDI control

Variable	Coefficient	Clustered SE	p-value
Trade openness	-0.001	(0.026)	0.959
ln(GDP per capita)	3.303*	(1.939)	0.098
Inflation	0.002	(0.010)	0.868
FDI inflows (% GDP)	0.128*	(0.072)	0.085
Observations	305		
Country clusters	35		
Adjusted R-squared	0.607		

Note. Dependent variable is asinh(public renewable-finance commitments).

Table 9. Sensitivity to wind-technology aggregation

Definition	Trade coefficient (cluster SE)	Two-sided p	Obs.
Conservative wind maximum	0.0027 (0.0268)	0.919	305
Additive wind categories	0.0022 (0.0269)	0.935	305

Note. Both specifications use two-way fixed effects; the alternative sums generic, onshore, and offshore wind labels as distinct categories. Neither estimate is statistically precise.

The alternative additive wind definition yields a trade coefficient of 0.0022 ($p = 0.935$), close to the conservative aggregation result; the null finding is not driven by this wind classification choice. The additive convention is a sensitivity check rather than proof that the IRENA taxonomy is mutually exclusive.

Table 10. Additional sensitivity checks for the trade-openness coefficient

Specification	β_1	Cluster SE	p-value	Obs.	Countries
Main two-way FE	+0.0027	0.0268	0.919	305	35
Exclude 2020–2021	+0.0029	0.0314	0.927	241	34
Exclude top 1% finance	+0.0070	0.0255	0.786	301	35
Lag trade one year	+0.0165	0.0271	0.546	275	35

Note. All estimates use two-way fixed effects and country-clustered standard errors. In the lag specification β_1 multiplies $T_{i,t-1}$; the other specifications use T_{it}. Samples vary with restrictions.
Source: authors' calculations.

Further sensitivity tests check whether the conclusion depends on a specific time period, a handful of large commitments or contemporaneous timing. Excluding 2020–2021 avoids the most direct pandemic-disruption years; deleting the top 1% of observed commitment amounts probes extreme finance observations; and using one-year-lagged openness probes delayed project approval. The lag test is not a valid instrument: openness and finance can both respond to persistent shocks. These exercises assess numerical stability rather than eliminate observational endogeneity.

4.4. Interpretation and limitations

Once missing finance cells are kept missing, the preferred model gives a trade coefficient close to zero with a wide confidence interval. The data do not reveal a precise association. This does not mean

that openness is economically irrelevant. The result concerns public commitments, not the combined private and public spending that creates clean-energy assets.

Across the additional specifications, the two-way fixed-effects trade coefficient remains statistically insignificant: the p-values are 0.927 without 2020–2021, 0.786 after omitting the largest 1% of finance entries and 0.546 with one-year-lagged trade. These numerical checks support a cautious "not detected" result rather than the stronger claim that openness has no meaningful influence under all conditions. Differences between public and private financing, and heterogeneous impacts across institutions and policy environments, remain plausible explanations.

Several limits matter for interpretation. The outcome leaves out private investment and records commitments rather than actual disbursements. Countries with observed finance data may differ from those with missing entries. Some wind categories may overlap, and their relationship cannot be resolved from this export. Large commitments, donor allocation, changing national policies and reverse causality also complicate the analysis. A longer panel or project-level records would be needed for stronger identification; no causal instrument is claimed here.

5. Conclusion and Policy Implications

This study investigated whether greater trade openness is associated with higher clean-energy investment in emerging markets. Rather than equating renewable capacity or energy consumption with investment, it examined a clearly delimited dependent variable: public renewable-energy finance commitments reported by the International Renewable Energy Agency (IRENA). The sample was drawn from the International Monetary Fund's emerging and developing economy classification, with a pre-specified 2023 GDP threshold of US$100 billion. Merging IRENA with World Bank annual indicators for 2014–2023 produced 305 complete country-year observations across 35 economies. The econometric analysis compared pooled ordinary least squares, country fixed effects and two-way fixed effects, with country-clustered standard errors.

The corrected findings do not establish that trade openness reliably raises public renewable-finance commitments. In the preferred country-and-year fixed-effects specification, the trade coefficient was positive but extremely small (0.0027 asinh units per percentage point of trade/GDP) and imprecise ($p = 0.919$). Its 95 percent confidence interval spans both negative and positive associations. Pooled and country-fixed-effects results likewise failed to identify a statistically distinguishable trade relationship. The primary hypothesis of a positive association is therefore not supported at conventional significance levels. This is not proof of an exactly zero effect: the interval leaves room for a range of economically relevant relationships.

The first policy-relevant lesson concerns measurement and data quality. The IRENA records distinguish reported finance amounts from missing observations; unrecorded amounts cannot be assumed to be genuine zero commitments. The emerging-market panel is thus conditioned on

observed finance data and may not represent all eligible economies equally. Public approvals can be concentrated in a few unusually large projects, creating a skewed distribution. The main specification uses an inverse hyperbolic sine transformation, documents technology aggregation rules and retains missing observations as missing. Sensitivity tests involving alternate wind definitions and the highest-valued finance observations do not overturn the statistically inconclusive trade result.

Trade may nevertheless matter through channels this outcome cannot capture. Imports of solar modules, turbines and related components can make project technology more accessible, while export networks can facilitate learning. But openness may also increase fossil-intensive output and energy demand. Moreover, public finance commitments respond to donor priorities, project preparation, regulatory approval and external financing conditions that differ from private investment decisions. A country may become more trade-intensive without receiving more recorded public renewable-finance commitments; conversely, a large public commitment need not immediately produce physical renewable infrastructure.

Governments should not assume that general trade liberalisation alone will bring in clean-energy finance. If equipment imports face costly delays or transaction costs, improving access to renewable technology is a reasonable step. It still needs to be backed by prepared projects, stable procurement rules, transparent tenders, reliable grids, creditworthy buyers and lenders able to support long-term projects. Better statistics are equally important: commitments, disbursements, private investment and completed capacity should be recorded separately. These distinctions would help policymakers assess whether trade integration leads to working assets, rather than finance that has only been announced.

Finally, the results are observational. Country and year effects remove time-invariant differences and shared shocks, not reverse causality, donor selection or time-varying reforms. The sample comprises 35 countries over at most ten years; residual-dependence checks are exploratory, and our data do not justify claims about a long-run cointegrating relationship. Additional specifications that exclude the pandemic years or use lagged trade also fail to identify a precise trade association. Future research should seek project-level investment and disbursement records, verify non-overlapping technology categories, and exploit credible policy reforms if causal identification is the objective. The most defensible conclusion is that trade openness may facilitate clean-energy finance, but the average association with public renewable-finance commitments cannot be reliably distinguished from zero in these data.

6. References

Antweiler, W., Copeland, B. R., & Taylor, M. S. (2001). Is free trade good for the environment? *American Economic Review, 91*(4), 877–908. https://doi.org/10.1257/aer.91.4.877

Hussain, J., Zhou, K., Muhammad, F., Khan, D., Khan, A., Ali, N., & Akhtar, R. (2021). Renewable energy investment and governance in countries along the Belt & Road Initiative: Does trade openness matter? *Renewable Energy, 180*, 1278–1289. https://doi.org/10.1016/j.renene.2021.09.020

International Monetary Fund. (2025). World Economic Outlook database: Groups and aggregates information. https://www.imf.org/en/Publications/WEO/weo-database/2025/April/groups-and-aggregates

International Renewable Energy Agency. (2026). IRENASTAT: Public investments (2023 million USD) by country/area, technology and year. https://pxweb.irena.org/pxweb/en/IRENASTAT/IRENASTAT__Finance/PUBFIN_2026_H2_PX.px/

Keller, W. (2004). International technology diffusion. *Journal of Economic Literature, 42*(3), 752–782. https://doi.org/10.1257/0022051042177685

Kim, K., & Kim, Y. (2015). Role of policy in innovation and international trade of renewable energy technology: Empirical study of solar PV and wind power technology. *Renewable and Sustainable Energy Reviews, 44*, 717–727. https://doi.org/10.1016/j.rser.2015.01.033

Murshed, M. (2020). Are trade liberalization policies aligned with renewable energy transition in low and middle income countries? An instrumental variable approach. *Renewable Energy, 151*, 1110–1123. https://doi.org/10.1016/j.renene.2019.11.106

World Bank. (2026). World Development Indicators. https://databank.worldbank.org/source/world-development-indicators

Zeren, F., & Akkuş, H. T. (2020). The relationship between renewable energy consumption and trade openness: New evidence from emerging economies. *Renewable Energy, 147*, 322–329. https://doi.org/10.1016/j.renene.2019.09.006

Zhang, M., Zhang, S., Lee, C.-C., & Zhou, D. (2021). Effects of trade openness on renewable energy consumption in OECD countries: New insights from panel smooth transition regression modelling. *Energy Economics, 104*, 105649. https://doi.org/10.1016/j.eneco.2021.105649